Validation and AI Use

Decision, version, and boundary

Decision/user: buy-side investment committee; watch/defer NVIDIA common stock. Valuation date: September 24, 2026. Currency: USD. The analysis uses final evidence available through the Q2 FY2027 Form 10-Q filed August 26, 2026 and the market reference recorded in the course work. Run `python assignments/project-1-nvidia/analysis.py` from the repository root.

Validation register

Claim/output	Failure mode	Test and expected direction	Actual result	Disposition
Enterprise-to-equity bridge	Claims omitted or signs reversed	Known-answer bridge: 1,000 + 100 - 250 - 20 - 10 = 820; 820 / 100 = 8.20	PASS	Accepted
Historical base	Opening balance sheet does not reconcile	FY2026 assets must equal liabilities plus equity at 206,803	PASS	Accepted
Forecast statements	Model hides an imbalance	Every forecast-year balance gap must be below 0.05m	PASS; maximum 0.0m	Accepted
Cash linkage	Cash is typed rather than calculated	Cash must remain above the 10,000m floor after cash-flow calculation	PASS; minimum headroom 117,283.8m	Accepted with model limitation
DCF monotonicity	DCF signs are wrong	9% WACC must produce a higher value than 11%; 4% terminal growth must exceed 2%	PASS; 226.98 > 167.82 and 219.31 > 173.52	Accepted
Bridge arithmetic	EV and equity result conflict	Equity must equal EV + 62,556 - 8,468	PASS	Accepted
Highest-risk failure	Terminal assumption dominates the decision	Record terminal-value share of EV and qualify outcome	77.0% in base case	Qualified: terminal sensitivity is material
Peer consistency	P/E is treated as EV or over-relied on	Price and annual GAAP diluted EPS must share currency/basis; one peer cannot be a range	PASS; AMD P/E is equity-value reference only	Qualified/rejected as decision value
AI output	AI decides peer selection	Compare AI suggestion to predeclared peer policy and primary business disclosures	AMD transferability concern accepted; no AI price is used	Qualified

Locked Changed-Input Record

The prediction below was supplied by the student before this rerun. Only WACC will change; the FCFF forecast, cash/securities, debt, terminal growth, and share basis stay fixed.

Precommit timestamp	Commit/snapshot	Material input	Old -> new value/units	Student's expected output direction	Expected decision effect

| 2026-10-16 13:11:02 EDT | pending prediction commit | Base WACC | 10.0% -> 11.0% | ,ÄúI think it will increase.,Äù | If value increases, the watch/defer conclusion would weaken; if it falls, it would strengthen. |

Before/after output	Actual decision effect	Prediction reconciliation	Failure diagnosis/no-change reason	Evidence path
193.14/share at 10.0% WACC -> 167.82/share at 11.0% WACC (-25.33/share), run 2026-10-16 13:16:44 EDT	Watch/defer strengthens because the higher required return lowers the base value farther below the 221.70 market reference.	Incorrect: the prediction expected an increase, but value decreased.	A higher WACC increases the discount rate applied to the same FCFF and terminal value, reducing their present values.	`analysis.py` / `visible_output.txt`

Material AI-use register

Tool	Material task	Independent check	Disposition
Codex	Organized existing NVIDIA research, model structure, documentation, and scripts	Student must inspect code, rerun it, verify filing values, and own the changed-input prediction	Assistive draft; human retains decision and validation ownership

Limitations and monitoring

The conclusion depends most on the durability of Data Center demand, gross-margin execution, reinvestment/capex, and a terminal value that is 77.0% of base EV. Monitor reported Data Center growth, gross margin, operating cash-flow conversion, capacity commitments, customer concentration, China restrictions, and evidence on share count/bridge classifications. Revisit on the next filing or if these facts materially depart from the model path.